

Semantic similarity in geographic information retrieval for decision making; [Similitud semántica en la recuperación de información geográfica para la toma de decisiones]

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ABSTRACT ORIGINAL

People use the similarity to store and retrieve information comparing new situations with similar experiences occurred in the past also for learning and concept formation. While equality comparison between two objects can be calculated by computers quickly and accurately similarity is a complex problem to calculate. Even still difficult to determine similarity plays a fundamental role in many applications such as decision-making systems, data mining and pattern recognition. The same applies to the spatial similarity in the processes of recovery and integration of spatial information. In this paper a methodology based on the semantic processing of geospatial information is proposed. It consists of five stages: conceptualization, synthesis, application processing, retrieval and management. It uses ontology of New Genetic Soil Classification of Cuba and applying the measure of semantic similarity Conceptual Distance (DIS-C) combined with an implementation model TDD (Topology-Direction-Distance) to restore and support the user in the selection of spatial scenes. As a case study is considered the town of San Jose de Las Lajas located in the province of Mayabeque in western Cuba. © 2014 AISTI.